SUBASH CHAKRAVARTHY

Data Engineer

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SUMMARY

Data Engineer with 2 years of experience building batch and streaming pipelines, data reliability frameworks, and analytical models in Snowflake and Databricks, including feature tables consumed by production ML models. Proficient in dbt, Apache Airflow, Apache Kafka, Debezium, and Terraform. M.S. in Analytics, Northeastern University (GPA 3.6), graduating May 2026.

TECHNICAL SKILLS

Languages & SQL: Python (OOP, pandas, PySpark), Scala (Spark jobs, pipeline development), SQL (window functions, CTEs, complex joins, query tuning, relational modeling), Spark SQL, C++

Pipelines & Orchestration: Apache Airflow (DAGs, retry logic, SLA monitoring, alerting), Apache Kafka (streaming ETL, sliding windows, watermarks), Debezium (log-based CDC, PostgreSQL/MySQL), Kafka Connect, Spark Structured Streaming, Lambda architecture, idempotent ingestion, ETL/ELT

Cloud & Warehousing: AWS (S3, Glue, EMR, Redshift, Lambda, DMS), GCP (BigQuery), Azure (Data Factory), Snowflake, Databricks, Delta Lake, Apache Iceberg, data lakehouse, Terraform (IaC)

Data Modeling & dbt: dbt (models, tests, snapshots, data contracts), Medallion architecture (Bronze/Silver/Gold), star schema, dimensional modeling, SCD Type 2, data catalog, data lineage, data governance

Quality & DevOps: Data reliability engineering, data observability, Great Expectations, schema drift detection, row-count reconciliation, CI/CD (GitHub Actions), Docker, Kubernetes, Git, MongoDB, Redis, PagerDuty, pgvector, feature store design, AI-assisted development (Claude Code, GitHub Copilot)

PROFESSIONAL EXPERIENCE

Radiustek Projects India Pvt. Ltd. — **Chennai, India**
Data Engineer — *Jan 2023 - Aug 2024*

- Built three PySpark and Scala streaming ETL jobs on Databricks to ingest 20GB+ of daily ERP and CRM data into a Snowflake data lakehouse following Medallion architecture with incremental loads and schema evolution, moving the analytics team from overnight batch runs to same-day data availability.
- Optimized Spark job runtime from approximately four hours to under three by reworking partition keys on high-cardinality join columns, switching broadcast joins to sort-merge, and caching recomputed DataFrames.
- Developed 10+ KPI models in dbt using SCD Type 2 snapshots and window functions for rolling metrics. Registered all assets in a data catalog to establish a single verified source of truth in Snowflake, resolving a six-month reporting conflict between the finance and operations teams.
- Implemented a Debezium and Kafka Connect CDC pipeline on two PostgreSQL source systems, reducing ingestion lag from eight hours to under five minutes, a reduction of over 99%.
- Engineered reusable feature-store tables (RFM, payment velocity, engagement signals) consumed directly by downstream ML models, keeping feature definitions consistent across analytics and ML pipelines.
- Built a data quality framework in Airflow with 15 automated checks per pipeline run covering schema drift, null thresholds, and row-count reconciliation, with PagerDuty alerts and Great Expectations lineage tracking. Pipeline failures dropped by 40% and weekly debug time fell from five hours to under one, an 80%+ reduction.

Emirates Transformers & Switchgear Co. — **Dubai, UAE**
Data Engineering Intern — *May 2021 - Jul 2021*

- Integrated assembly logs and supply-chain records from four source systems using Python, Scala, and SQL. Idempotent ingestion patterns eliminated duplicate records and cut daily aggregation time by approximately 90 minutes, freeing the operations analyst to focus on same-day exception handling instead of manual dedup.
- Standardized field names, data types, and schemas across production, inventory, and logistics datasets, enabling same-day reporting without manual consolidation.

PROJECTS

Operational Data Warehouse: Batch ELT Platform *PySpark · Scala · Snowflake · Airflow · dbt · Great Expectations · GitHub Actions*

- Developed a batch ELT pipeline in Python and Scala that processes 150,000+ daily transactions into Snowflake across four dbt star-schema tables, enforcing 12 data-contract tests via Great Expectations. Ad-hoc analyst queries that previously took 45 minutes now complete in under three, a reduction of over 90%.
- Structured Airflow DAGs with exponential-backoff retries and SLA breach alerts, and deployed via GitHub Actions with automated dbt test gating on every merge.

Real-Time CDC + Streaming Platform *Python · Scala · Debezium · Kafka · Spark Streaming · Delta Lake · Docker · dbt*

- Built a CDC pipeline using Debezium and Kafka Connect to stream row-level PostgreSQL changes through a Lambda-architecture platform handling 80,000+ events per day. The platform uses five-minute sliding windows and watermark-based late-arrival handling, maintaining end-to-end latency within a 10-minute SLA.
- Implemented exactly-once delivery through deduplication keys and Delta Lake ACID transactions, with dbt transformations enforcing data quality at every layer. Reporting latency improved from next-day batch to under 15 minutes.

CERTIFICATIONS

Data Engineering Professional Certificate (Snowflake) · Microsoft Azure Essentials Professional Certificate (Microsoft/LinkedIn), 2026

EDUCATION

Northeastern University — **Boston, MA**
M.S. in Analytics - GPA: 3.6/4.0 — *May 2026*
Coursework: Data Management & Big Data, Database Management Systems, Data Mining, Data Governance

Vellore Institute of Technology — **Vellore, India**
B.Tech., Electronics & Communication Engineering — *May 2023*